THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · MEDIUM LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A student models a falling ball ignoring air resistance. For which object is the model least accurate? 1

(a) a feather **(b)** a steel bolt **(c)** an iron ball **(d)** a stone

2. A solution is prepared by dissolving 30 g of sugar in 120 g of water. The mass percentage is: 1

(a) 30 per cent **(b)** 25 per cent **(c)** 12.5 per cent **(d)** 20 per cent

3. In an experiment 5.6 g of CaO reacts with 4.4 g of CO2 to form CaCO3. The mass of CaCO3 formed is: 1

(a) 11 g **(b)** 1.2 g **(c)** 9.6 g **(d)** 10 g

4. An atom has the configuration 2, 8, 6. The number of protons is: 1

(a) 8 **(b)** 16 **(c)** 14 **(d)** 6

5. A car starting from rest has acceleration 3 m/s squared. Its velocity after 4 s is (m/s): 1

(a) 1 m/s **(b)** 12 m/s **(c)** 7 m/s **(d)** 6 m/s

6. A force of 24 N acts on a body and produces an acceleration of 4 m/s squared. The mass is: 1

(a) 96 kg **(b)** 6 kg **(c)** 20 kg **(d)** 0.17 kg

7. A force of 20 N acts at right angles to the displacement of 5 m. The work done is: 1

(a) 4 J **(b)** 100 J **(c)** zero **(d)** 25 J

8. The frequency of a wave is 500 Hz and the speed is 340 m/s. The wavelength is: 1

(a) 170000 m **(b)** 0.68 m **(c)** 0.0015 m **(d)** 1.47 m

9. A plant cell is placed in a solution and the cell contents shrink away from the wall. The solution is: 1

(a) hypertonic **(b)** hypotonic **(c)** pure water **(d)** isotonic

10. Which tissue would be most affected in a plant if the cambium is removed? 1

(a) gas exchange **(b)** increase in girth
(c) increase in length **(d)** water absorption

11. An organism is unicellular, eukaryotic and has chloroplasts and a flagellum. It belongs to: 1

(a) Plantae **(b)** Protista **(c)** Fungi **(d)** Monera

12. A farmer wants to grow a seedless banana plant. The method he uses is: 1

(a) seed sowing **(b)** cross-pollination
(c) grafting only **(d)** vegetative propagation by suckers

13. The distance from the Earth to the Sun is 1.5 x 10^11 m and the speed of light is 3 x 10^8 m/s. The time taken by sunlight is: 1

(a) 50 s **(b)** 5 s **(c)** 500 s **(d)** 5000 s

14. A pendulum model assumes no friction. The actual pendulum: 1

(a) slowly stops swinging **(b)** swings forever
(c) stops at once **(d)** speeds up

15. A mixture of two liquids X (b.p. 78 degrees Celsius) and Y (b.p. 118 degrees Celsius) is best separated by: 1

(a) filtration **(b)** sublimation
(c) fractional distillation **(d)** crystallisation

16. The formula of aluminium sulphate is: 1

(a) $Al(SO_4)_3$ **(b)** $Al_3(SO_4)_2$ **(c)** $AlSO_4$ **(d)** $Al_2(SO_4)_3$

17. An element has 3 shells and 2 valence electrons. Its atomic number is: 1

(a) 14 **(b)** 12 **(c)** 10 **(d)** 8

18. A body moves along a straight line with speed 4 m/s for 10 s and 6 m/s for the next 10 s. The average speed is: 1

(a) 10 m/s **(b)** 4 m/s **(c)** 6 m/s **(d)** 5 m/s

19. A person weighs 600 N on the Earth. On a planet with g one-third of the Earth's, the weight is: 1

(a) 600 N **(b)** 1800 N **(c)** 300 N **(d)** 200 N

20. A lamp of 60 W is used for 5 hours daily for 30 days. The energy consumed in kWh is: 1

(a) 300 **(b)** 1.8 **(c)** 90 **(d)** 9

21. Sound of two different tuning forks has frequencies 256 Hz and 512 Hz. The wavelength is: 1

(a) zero **(b)** greater for the 256 Hz fork
(c) greater for the 512 Hz fork **(d)** equal

22. A cell has 8 chromosomes. After mitosis each daughter cell has: 1

(a) 2 **(b)** 8 **(c)** 16 **(d)** 4

23. A section of an animal tissue shows flat cells with a single layer. This is likely: 1

(a) cuboidal epithelium **(b)** bone
(c) columnar epithelium **(d)** squamous epithelium

24. A fish differs from an amphibian in having: 1

(a) lungs **(b)** gills throughout life
(c) moist skin **(d)** a three-chambered heart

25. A flower has stamens and a carpel. It is a: 1

(a) bisexual flower **(b)** female flower **(c)** unisexual flower **(d)** male flower

26. A 1 cm rise of the sea level due to melting ice would mostly affect: 1

(a) deserts **(b)** coastal low-lying areas
(c) mountain tops **(d)** polar caps only

27. The density of an object is 2.5 g per cubic cm. In kg per cubic metre this is: 1

(a) 25000 **(b)** 2500 **(c)** 250 **(d)** 25

28. A student has a mixture of sand, salt and camphor. The correct sequence of separation is: 1

(a) filtration, sublimation, evaporation
(b) evaporation, sublimation, filtration
(c) sublimation, dissolving and filtering, evaporation
(d) dissolving, sublimation, evaporation

29. Which of these pairs forms an ionic compound? 1

(a) C and H **(b)** N and H **(c)** Mg and Cl **(d)** C and O

30. Chlorine has isotopes of mass 35 u (75 per cent) and 37 u (25 per cent). The average atomic mass is: 1

(a) 36 u **(b)** 37 u **(c)** 35 u **(d)** 35.5 u

31. A stone thrown vertically up returns to the thrower's hand after 6 s. The velocity of throw is (g = 10): 1

(a) 30 m/s **(b)** 10 m/s **(c)** 15 m/s **(d)** 60 m/s

32. A block of mass 4 kg rests on a rough table. The least force of 12 N makes it just move. Static friction is: 1

(a) 4 N **(b)** 12 N **(c)** 0 **(d)** 40 N

33. In a lever, the effort arm is 40 cm and the load arm is 10 cm. The MA is: 1

(a) 4 **(b)** 0.25 **(c)** 50 **(d)** 30

34. Two musical instruments produce notes of the same pitch and loudness but you can tell them apart. This is due to differing: 1

(a) quality or timbre **(b)** speed
(c) amplitude **(d)** frequency

35. The organelle which is found in all eukaryotic cells but whose number is highest in actively respiring cells: 1

(a) cell wall **(b)** vacuole **(c)** nucleus **(d)** mitochondria

36. A person can bend the arm at the elbow only in one plane. This joint is a: 1

(a) ball and socket **(b)** hinge joint **(c)** pivot joint **(d)** fixed

37. The correct sequence of taxonomic ranks for the tiger from the highest is: 1

(a) Animalia, Chordata, Mammalia, Carnivora, Felidae, Panthera, tigris
(b) Animalia, Mammalia, Chordata, Felidae, Carnivora, Panthera
(c) Panthera, Felidae, Carnivora, Mammalia
(d) Chordata, Animalia, Mammalia, Carnivora

38. The number of chromosomes in a human zygote is: 1

(a) 23 **(b)** 46 **(c)** 92 **(d)** 44

39. In the water cycle, the Sun provides energy for: 1

(a) condensation **(b)** run-off **(c)** precipitation **(d)** evaporation

40. A runner covers 100 m in 12.5 s. The estimated speed is: 1

(a) 80 m/s **(b)** 8 m/s **(c)** 12.5 m/s **(d)** 1.25 m/s

41. How many grams of salt are present in 250 g of a 4 per cent salt solution? 1

(a) 25 g **(b)** 10 g **(c)** 4 g **(d)** 40 g

42. Ammonia contains 82.35 per cent nitrogen. In 34 g of ammonia the mass of nitrogen is: 1

(a) 6 g **(b)** 17 g **(c)** 28 g **(d)** 14 g

43. An element has atomic number 19. The number of valence electrons is: 1

(a) 1 **(b)** 9 **(c)** 2 **(d)** 8

44. A body starts from rest with acceleration 2 m/s squared. The distance covered in 10 s is: 1

(a) 200 m **(b)** 100 m **(c)** 10 m **(d)** 20 m

45. A body of mass 3 kg moving at 4 m/s is stopped in 2 s. The average force is: 1

(a) 6 N **(b)** 3 N **(c)** 24 N **(d)** 12 N

46. A skier of mass 60 kg slides down from rest through a vertical drop of 20 m. The speed at the bottom if friction is negligible (g = 10) is: 1

(a) 20 m/s **(b)** 400 m/s **(c)** 40 m/s **(d)** 10 m/s

47. A sound wave has a frequency of 2000 Hz and a speed of 340 m/s. The number of waves in a 3.4 m length is: 1

(a) 2 **(b)** 10 **(c)** 20 **(d)** 40

48. Dividing cells of a root tip are best seen under the microscope in the: 1

(a) growing region **(b)** mature root **(c)** root hair zone **(d)** root cap

49. A stem with a girth increasing yearly has active: 1

(a) root cap **(b)** stomata **(c)** apical meristem **(d)** vascular cambium

50. A plant is described as having vascular bundles scattered in the stem, parallel venation and fibrous roots. It is a: 1

(a) dicot **(b)** pteridophyte **(c)** monocot **(d)** gymnosperm

END OF QUESTION PAPER **Total: 50 Marks**

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